Explanation Document

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1. Approach

This problem was formulated as a **hierarchical multi-label classification** task. Because a single consumer post often contains multiple themes, traditional single-label classifiers are insufficient.

- **Taxonomy Extraction:** I first parsed `Training.xlsx` to programmatically extract the exact hierarchy (17 Level 1 parent categories and 42 nested Level 2 subcategories).
- **Prompt Engineering:** I utilized in-context learning (few-shot prompting) by providing the LLM with the strict taxonomy constraints, rules for multi-label extraction, and 2-3 representative examples from the training set.
- **Schema Enforcement:** The model was constrained to output structured JSON, ensuring that every predicted Level 2 factor strictly belonged to its mapped Level 1 parent. The JSON arrays were then flattened to match the requested "one row per mapping" output format.

2. Models Selection

- **Llama 3.1 8B (Considered):** While fast and cost-effective, compact 8B models occasionally hallucinate tags outside the provided taxonomy or break JSON syntax under strict constraints. Also it was not under free tier of groq.
- **openai/gpt-oss-120b (Selected):** Accessed via the Groq API, this 120-billion parameter model was chosen for its superior semantic reasoning and strict adherence to JSON formatting. It successfully differentiated between highly nuanced categories (e.g., *Sensory* vs. *Feel*) and flawlessly handled multi-label outputs.

3. Techniques, Tools, and Data Used (Model "Training")

Due to the small dataset size, traditional model weight updating (fine-tuning) was bypassed in favor of **prompt-based in-context learning**.

- **Data Used:** `Training.xlsx` was used to map the allowed taxonomy and source high-quality few-shot examples demonstrating single-label and multi-label scenarios.
- **Tools:** Python, Pandas (for data manipulation), Scikit-learn (for evaluation metrics), and the Groq API (for inference).
- **Technique:** To adhere to Groq's free-tier rate limits (8,000 Tokens-per-Minute), the inference loop utilized strategic pacing (`time.sleep`) and exponential backoff, ensuring 100% completion of the test set without API failures.

4. Output Accuracy

Because `Process.xlsx` lacks ground-truth labels, accuracy was evaluated using a 20% stratified hold-out validation split on the `Training.xlsx` dataset.

- **Level 1 Micro F1 Score:** 0.7228
- **Hierarchical Micro F1 Score (Level 1 + Level 2):** 0.6154
- **Subset Exact Match Accuracy:** 33.33%